# Metal-specific anticancer activity does not transfer across metals: a cross-metal leave-one-metal-out benchmark on MetalCytoToxDB

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**Abstract**

Ligand-based activity models are routinely assumed to transfer across chemical scaffolds. We ask whether the same assumption holds when the variable that changes is the *metal*, not the scaffold. This cross-metal axis is not benchmarked in either the ligand-based or the transition-metal literature, so we make it the primary measurement. Using a deduplicated cohort drawn from MetalCytoToxDB (5,271 rows, five precious/platinum-group metals Ru/Ir/Os/Rh/Re), we measure within-metal 5-fold cross-validated ROC-AUC against cross-metal leave-one-metal-out (LOMO) ROC-AUC under a single shared scorer. We find a systematic *collapse*: within-metal AUC is 0.70–0.95 for Ru, Ir, Rh and Re (Os's within-metal signal does not resolve), versus 0.44–0.62 when the same model predicts a held-out metal. **MEASURED**. A label-permutation null, Holm-corrected over the five-metal family, shows that no metal's cross-metal AUC is significantly *below* chance under the fingerprint scorer — only Ru is significantly above it there, and Rh/Os/Re are unresolvable at $n = 36$–$74$. A message-passing backbone reproduces the collapse for Ru, Ir and Rh but recovers Re (0.737, Holm $p = 0.010$); under the same correction the backbone leaves only Re and Ru significant, and Ir's uncorrected below-chance reading ($p = 0.038$) does not survive. A same-metal Bemis–Murcko scaffold-split control costs 0.10–0.12 on the two metals where it resolves, roughly half of the collapse, so the metal axis is the largest single shift but not the only one. **MEASURED**. We then ask what repairs the collapse: static metal conditioning — a metal token, a descriptor embedding, or FiLM — does *not*, but few-shot adaptation, fine-tuning on $K$ labelled ligands from the held-out metal, recovers it monotonically, by +0.058 at $K$=20 and +0.091 at $K$=40 (Holm $p < 0.004$). **MEASURED**. We also test whether tmQM pre-training of the graph encoder transfers to the activity head, on two metals, and report a second honest negative: on a genuine scaffold split no Holm-corrected effect is significant — Ru's paired effect is −0.0094 (t-based 95% CI [−0.0406, +0.0218]) and Ir's suggestive +0.0486 does not clear the paired $t$-test (one-sided $p = 0.068$) — reversing a +0.0181 that had been measured on a degenerate, molecule-level splitter. **MEASURED**. Against the published protocol on both of its axes: on its *within-metal* axis our fingerprint-plus-XGBoost scorer exceeds the published 0.73 under both tie rules and every seed (0.8237–0.8597, +0.09 to +0.13), while Ru is at parity and decided by the tie rule, and we report the first within-metal values for Rh, Os and Re, which the published work does not cover; on its *DOI-split* replication our D-MPNN scores $0.7717 \pm 0.0380$ against the published LightGBM $0.810 \pm 0.024$ — 0.0383 below, on a cohort 25× wider (591 vs 23 cell lines). Running the published model class on our identical cohort and fold partition scores $0.7677 \pm 0.0368$, so the residual gap is cohort scope, not model class. **MEASURED**. We conclude that the metal identity is a first-class distribution shift that the ligand-based QSAR literature has no routine analogue for, that cross-metal transfer is a benchmark axis worth reporting explicitly, and that few-shot adaptation to the target metal — not metal conditioning — is the practical fix.

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# 1 Introduction

Ligand-based activity prediction rests on an assumption that is rarely stated explicitly: that the chemical features which determine whether a compound is active are carried by the organic scaffold, and therefore that a model trained on one set of compounds will rank a new set of compounds sensibly even if the scaffolds differ. This assumption is the reason scaffold splits (1) are treated as the honest evaluation for molecular property models, and the reason transfer across organic chemical space is considered a solved-enough problem to build benchmarks on (2).

For organometallic and coordination compounds the assumption is weaker in a specific, testable way. Two complexes can share an identical ligand set and differ only in the coordinated metal, and that single change can move the measured anticancer activity by orders of magnitude. The metal is not a substituent; it sets the geometry, the accessible oxidation states, the hydrolysis rate, and the DNA-binding mode. If a ligand-based model has learned "which scaffolds are toxic", it should transfer across metals. If it has learned "which metal–ligand combinations are toxic", it should not.

We test this directly. Using the MetalCytoToxDB cohort of precious- and platinum-group anticancer complexes (3), we evaluate the *same* model and the *same* scorer under three regimes: within-metal 5-fold cross-validation, a same-metal Bemis–Murcko scaffold-split control, and cross-metal leave-one-metal-out (LOMO) transfer. The within- and cross-metal regimes differ in more than the held-out metal — their training sets, sample sizes and label priors all change — so we use the scaffold control to measure how much of the within-to-cross drop is ordinary novel-scaffold generalisation and how much is specific to the metal axis. Neither the ligand-based nor the transition-metal literature reports this cross-metal measurement (§2), so we treat it as the primary object of study rather than an incidental split.

Our contributions are:

1. **A measured collapse, its decomposition, and its robustness.** Within-metal ROC-AUC is 0.70–0.95 for Ru, Ir, Rh and Re (Os's within-metal signal does not resolve), but cross-metal LOMO AUC falls to 0.44–0.62 for the same model. A same-metal Bemis–Murcko scaffold-split control costs 0.10–0.12 on the two metals where it resolves, so roughly half of the collapse is ordinary novel-scaffold generalisation and the metal axis is the largest single remaining shift, not the only one. The collapse is not an artefact of the scorer: it reproduces under an independent message-passing backbone on the same cohort and split, which additionally recovers Re (0.737, Holm $p = 0.010$) where the fingerprint probe scores 0.443 — so we report Re as representation-dependent. Under a Holm-corrected label-permutation null the fingerprint scorer leaves only Ru significantly above chance, and the backbone leaves only Re and Ru significant: Ir's below-chance reading is significant only uncorrected ($p = 0.038$) and does not survive the correction. We argue metal identity is a distribution shift with no routine organic analogue, and that it should be reported as a benchmark axis in its own right. **MEASURED**. §4.

2. **No significant pretraining transfer, measured on two metals.** A graph encoder pretrained on the tmQM transition-metal corpus (4) does not raise the activity head on a genuine scaffold split. On Ru the paired delta is $-0.0094$ (t-based 95% CI $[-0.0406, +0.0218]$, $t(4) = -0.84$, 2/5 seeds positive); on Ir it is a suggestive $+0.0486$ that does not clear the paired $t$-test (3/5 seeds positive, one-sided $p = 0.068$, CI includes zero). An earlier $+0.0181$ was measured on a degenerate, molecule-level splitter and does not survive the fix. Both arms print the tensor-load count, so a silent no-op is excluded; the null is a null of the effect, not of the measurement. **MEASURED**. §4.

3. **A within-metal win and a cross-cohort honest negative.** On the source paper's own within-metal axis our fingerprint-plus-XGBoost scorer exceeds the published 0.73 under both tie rules and every seed (0.8237–0.8597); Ru is at parity, decided by the tie rule; and we report the first within-metal values for Rh, Os and Re, which the source paper does not cover. Under their DOI-split 10-fold cross-validation our model scores $0.7717 \pm 0.0380$ against the published LightGBM $0.810 \pm 0.024$: 0.0383 below, with 3 of 10 folds at or above 0.81, on a cohort 25× wider. We report both the win and the gap rather than burying either. **MEASURED**. §4.

4. **A method, and a clean negative for the obvious alternative.** Static metal conditioning — a metal token, a metal-descriptor embedding, or a FiLM scale-and-shift — does not repair the collapse: all three leave the cross-metal mean at or below the unconditioned baseline (FiLM $-0.0418$, Holm $p = 0.984$). What does work is few-shot adaptation. Fine-tuning on $K$ labelled ligands from the held-out metal recovers AUC monotonically, by $+0.058$ at $K$=20 and $+0.091$ at $K$=40 (Holm $p < 0.004$, ten seeds). The collapse is therefore not a missing-input problem but a target-distribution problem, and a handful of target-metal labels is the cheap fix. **MEASURED**. §4.4.

The paper is organised as follows. §2 positions the work against ligand-based QSAR and transition-metal machine learning. §3 defines the cohort, the two evaluation regimes, and the model. §4 reports the four claims. §5 gives the sensitivity analyses that bound the conclusions (tie rule, deduplication, pretraining A/B). §6 states what the study does *not* establish, including a cohort-truncation effect we found in a related protocol and the small per-metal sample sizes that make part of the result unresolvable.

## 2 Related work

Our study sits at the intersection of three literatures: ligand-based activity prediction and the transfer assumption behind its evaluation, machine learning for transition-metal complexes, and evaluation protocols for distribution shift. We take each in turn and close with the specific gap we address.

### 2.1 Ligand-based activity prediction and its transfer assumption

Modern ligand-based activity prediction is dominated by message-passing graph neural networks (5, 6). The directed message-passing neural network (D-MPNN) of Yang et al. (7) remains a standard backbone, and its reference implementation (8) is the de-facto baseline on MoleculeNet (2). The family has since been extended along three axes that matter here: attention over atom environments (9), contrastive self-supervision (10), and large-scale pretraining on chemical language (11). These sit inside a broader programme of machine learning for chemistry (12, 13) and molecular design (14, 15), whose activity-prediction applications now extend to prospectively validated discoveries (16).

What all of these share is an evaluation convention: a *scaffold split*, in which the Bemis–Murcko scaffold (1) of every test compound is absent from training, so that the reported AUC is a transfer estimate over organic chemical space. The convention encodes an assumption — that scaffold novelty is the hard case. Our question is orthogonal. We hold the evaluation machinery fixed and vary a different axis: whether the *metal centre* of a coordination compound was observed during training. A scaffold split of metal complexes does not control for this, because the same organic scaffold can appear on several metals.

## 2.2 Machine learning for transition-metal complexes

Transition-metal complexes break several assumptions that organic benchmarks rely on. Molecular graphs are ambiguous about bond order at the metal, valence is variable, and the electronic structure that drives activity is not fully captured by a 2D graph; the machine-learning approaches developed for this chemical space are surveyed by Nandy et al. (17). Concretely, feature-selection studies for first-row transition-metal complexes (18) and active-learning searches over spin-crossover space (19) showed that quantum-chemical targets for coordination compounds are learnable from structure alone.

Two strands of that work bear directly on our design. The first is pretraining on quantum-chemistry corpora. The tmQM corpus (4) supplies quantum geometries and properties for ∼86k transition-metal complexes, and pretraining on it is a natural analogue of the large-scale pretraining that lifted organic property prediction. It is one of a family of datasets that made such pretraining credible — from QM9 (20) and ANI-1 (21) to the Open Catalyst Project (22) and crystal-graph models (23) — and the tmQMg extension adds excited-state properties for a further 74k complexes (24). The second is metal-aware descriptors. Orsi and Frei (25) build a universal metal fingerprint from electron configuration, which is exactly the kind of inductive bias a shared ligand-only head lacks.

The metallodrug context that motivates the activity endpoint begins with cisplatin (26) and its cellular pharmacology (27), and has since broadened to ruthenium (28) and organometallic (29) anticancer complexes; the drug-design perspective is surveyed by Ndagi et al. (30) and the medicinal-inorganic framing by Mjos and Orvig (31).

On the activity side, Krasnov et al. (3) assembled MetalCytoToxDB, a curated database of anticancer transition-metal complexes spanning Ru, Ir, Rh, Re and Os, and reported per-metal binary activity models (Ru 0.81, Ir 0.73 AUC) trained on Morgan fingerprints with gradient-boosted trees under a DOI-grouped 10-fold split. That work establishes both the data and the protocol we adopt, and it is the baseline against which we measure our honest negative in §4.

## 2.3 Evaluation protocols and distribution shift

Our finding is a statement about evaluation protocol, so it also belongs to the literature on how molecular models are benchmarked and where they fail. Domain shift in molecular machine learning is usually studied as a shift in scaffold, size, or assay, and is benchmarked explicitly by datasets such as DrugOOD (32); the general problem is studied as domain generalisation, for which comprehensive surveys exist (33, 34), and the closely related covariate-shift setting has its own estimators and validation schemes (35). The broader graph-learning-for-drug-discovery toolkit is surveyed by Zhang et al. (36), and the question of when a deep QSAR model can be trusted outside its training distribution — its applicability domain — is examined by Liu et al. (37).

The shift we study is not of that kind. It is a change in the identity of a single atom whose chemistry reorganises the whole molecule, so the test distribution is not a novel point in the same chemical space but a different conditional relationship between the ligand features and the label. We are not aware of an organic benchmark that isolates this effect, which is precisely why we treat it as a finding rather than an implementation detail.

## 2.4 What is missing

Across both the ligand-based and the transition-metal literature, transfer is reported *within* a metal or across organic chemical space. A per-metal model answers "how well can we predict activity for this metal"; it does not answer "does what we learn about one metal help with another". No

study in either line publishes a cross-metal leave-one-metal-out measurement, and none uses a same-metal scaffold split as the control that separates the metal axis from ordinary novel-scaffold generalisation. We make cross-metal transfer the primary measurement, and we report it under two independent scorers so that the conclusion does not rest on one model class.

# 3 Method

## 3.1 Cohort and deduplication

We use MetalCytoToxDB (3), restricted to the five metals it covers: Ru, Ir, Rh, Re and Os. Cytotoxicity is binarised from the dark IC50 with a 10 $\mu$M activity threshold; rows below a 0.01 $\mu$M noise floor are dropped.

Measurement-level duplicates are collapsed to one row per unique ligand: for every ligand the label is the majority over its measurements, and ties are broken *toward active* (the predicate is sum $\times$ 2 $\geq$ $n$). Ties going to active is the conservative choice for a cytotoxicity endpoint and is the rule the released code actually implements. Duplication is severe and uneven: Ru has a 65.8% duplication rate (3,668 $\rightarrow$ 1,255 unique ligands), Ir 57.0%, Os 58.6%, Rh 52.8%. Because the same organic ligand can be measured against several metals, a raw row split leaks ligands across the train/test boundary; the LOMO protocol below removes that leak explicitly.

## 3.2 Three evaluation regimes

We hold the cohort and the scorer fixed and vary only the split.

**Within-metal.** For each metal, 5-fold cross-validation over that metal's unique ligands. This is the setting the source paper reports, and it answers "how well can activity be predicted for a metal we have data on".

**Within-metal scaffold (control).** For each metal, the same 5-fold CV but with whole Bemis–Murcko scaffolds held out, so train and test share no scaffold (1). This measures the ordinary novel-scaffold generalisation penalty on this cohort and is the reference the cross-metal number must beat before a metal-specific effect can be claimed.

**Cross-metal (LOMO).** For each held-out metal $m$, train on the unique ligands of the other four metals and test on $m$. The leakage rule is strict: any training ligand that also occurs in *any* other metal, including the held-out one, is dropped from the train pool before fitting. This makes the test set ligand-disjoint from training by construction.

The three regimes differ in more than the held-out metal: their training sets, sizes and label priors all change. We therefore do not read the raw within-to-cross drop as a pure metal effect. We decompose it into the novel-scaffold penalty (within $-$ scaffold) and the residual metal effect (scaffold $-$ cross), and report both in Table 1.

Per-metal cohorts are small — Ru 1,255 unique ligands, Ir 534, Re 74, Rh 51, Os 36 — so we report each metal separately and never quote a pooled cross-metal mean without the per-metal table. Ru is 64% of the deduplicated cohort, so a pooled number is largely a Ru number.

## 3.3 Models

**Probe scorer (Claim 1).** Morgan fingerprints (radius 2, 2048 bits) with XGBoost. This is the standard ligand-based baseline and keeps the within/cross comparison free of any neural architecture confound: the collapse, if real, must appear for a fingerprint model too.

**D-MPNN backbone (Claims 2–4).** A directed message-passing neural network (7) with a shared encoder and per-metal activity heads, trained with a Wasserstein multi-task objective. Hidden width 128, depth 3, batch 64, learning rate $10^{-3}$, 30 epochs, seeds $\{0, \ldots, 4\}$. The activity loss is binary cross-entropy with a per-metal positive weight; no sampler is used. The MoleculeNet backbone control (§4.5) uses the same encoder at hidden width 512, depth 4 — the configuration that maximises the organic-property control — and is otherwise trained identically.

**Scorer.** ROC-AUC on held-out ligands, macro-averaged over seeds. We also report PR-AUC because the positive rate varies across metals.

### 3.4 Metal conditioning and few-shot adaptation

To ask what repairs the collapse (§4.4) we hold the backbone, cohort and ligand-disjoint LOMO split fixed and change only how the target metal enters. In the *static conditioning* arms the metal identity is supplied at training time as (i) a metal token concatenated to the graph embedding, (ii) a learned embedding of a metal-descriptor vector, or (iii) a FiLM scale-and-shift $h \mapsto (1+\gamma) \odot h + \beta$ on the pooled readout, where $(\gamma, \beta)$ is produced from the metal descriptor. In the *few-shot* arm the zero-shot model is frozen and then fine-tuned for 20 epochs on $K$ labelled support ligands drawn from the held-out metal, and scored on a disjoint query set from the same metal, so the reported delta is paired against the same run's zero-shot query AUC. We sweep $K \in \{5, 10, 20, 40\}$ over ten seeds and all five metals and correct across the family with Holm–Bonferroni. The support and query sets are ligand-disjoint, so no query ligand is seen during adaptation.

### 3.5 tmQM pretraining (A/B)

The encoder is either initialised from a checkpoint pretrained on the tmQM transition-metal corpus (4), or from a random seed. The two arms are identical in every other respect: same cohort, same Bemis–Murcko scaffold split, same 30 epochs, same five seeds, and the seed varies only the model initialisation. We run the A/B on two metals (Ru, Ir) rather than one. Both arms log the number of backbone tensors loaded from tmQM, so an arm that silently failed to load would be visible rather than masquerading as a successful transfer. The measured quantity is the *paired* per-seed delta, not the difference of means, and it is tested with a paired one-sided $t$-test over the five seeds (a 5-seed bootstrap is anti-conservative; see §3.7).

### 3.6 Replication of the published protocol

For the head-to-head against Krasnov et al. (3) we adopt their criteria rather than our own: measurement-level dedup on (`SMILES_Ligands`, `Cell_line`, `Time(h)`) keeping the first record, a 76-name non-cancer cell-line exclusion applied by exact stripped-uppercase match, and the paper's primary evaluation scheme, a DOI-grouped 10-fold cross-validation in which all rows sharing a DOI fall in the same fold (DOI-disjointness verified). Our cohort is larger than theirs — 15,799 rows / 4,793 complexes / 591 cell lines, against 11,294 / 4,399 / 23 — so the comparison is a cross-cohort estimate, labelled as such, not a reproduction. The like-for-like numbers are the two same-cohort rows in §4, which hold our cohort and fold partition fixed and vary only the model class; the comparison to the published 0.810 is cross-cohort and labelled as such.

### 3.7 Statistics

The cross-metal transfer penalty (within − cross, and scaffold − cross) is reported with a 95% percentile bootstrap confidence interval (2,000 resamples of the held-out ligands) (38), computed

Table 1: Within-metal versus cross-metal transfer, same cohort and same scorer. Within-metal is the 5-seed mean over the 5-fold CV (seed range in brackets). Scaffold is a same-metal Bemis–Murcko scaffold split (single seed 42); cross is the ligand-disjoint LOMO evaluation (single seed 42). Net metal = scaffold − cross, i.e. the collapse with the ordinary novel-scaffold penalty removed. The permutation verdict is a label-permutation null on the fixed cross-metal ranking (1,000 shuffles, seed 42), Holm–Bonferroni-corrected over the five-metal family; only Ru survives the correction. **MEASURED**: `metrics/by_metric/metal_activity_transfer_seed_stability.json`, `metal_activity_transfer_scaffold.json`, `metal_activity_transfer_permutation_null.json`, `metal_activity_transfer_significance.json`.

| Metal | Within (5-fold) | Scaffold | Cross (LOMO) | Net metal | Verdict |
|---|---|---|---|---|---|
| Ru | 0.8271 [0.8128, 0.8363] | 0.7338 | 0.6179 | +0.116 | above (sig) |
| Ir | 0.8430 [0.8295, 0.8597] | 0.7087 | 0.5276 | +0.181 | in null |
| Rh | 0.6974 [0.6429, 0.7381] | 0.4690 | 0.5190 | −0.050 | in null |
| Os | 0.4960 [0.3133, 0.5867] | 0.4900 | 0.4950 | −0.005 | in null |
| Re | 0.9480 [0.9083, 0.9819] | 0.8465 | 0.4429 | +0.404 | in null |

on the fixed seed-42 measurement. The pretraining delta in Table 2 is a per-seed *paired* quantity, so seed-to-seed variance cancels; it is tested with a paired one-sided $t$-test over the five seeds. A percentile bootstrap on five deltas is recorded in the artifact for reference but is anti-conservative, so the family correction is not based on it. Because we test five held-out metals against a label-permutation null, we control the family-wise error rate over that family of five with a Holm–Bonferroni step-down correction (39) and report which metals survive it; the uncorrected per-metal verdicts are shown alongside so a reader can see both. All p-values are two-sided.

### 3.8 Reproducibility

Every number in this paper traces to a JSON artifact under `metrics/by_metric/`; the mapping from claim to file is given at each table and figure. Scripts, configs and cohorts are versioned in the same repository. We follow a strict MEASURED/PROJECTED discipline: no projected value appears in a results table.

## 4 Results

### 4.1 Claim 1: metal-specific activity does not transfer across metals

Table 1 reports, for the same fingerprint-plus-XGBoost scorer and the same deduplicated cohort, within-metal 5-fold AUC against cross-metal LOMO AUC. Figure 1 shows the same data as a transfer matrix.

Three observations follow. First, the collapse is large and consistent: for Ru, Ir, Rh and Re the cross-metal AUC is 0.18–0.51 lower than within-metal (the raw penalties are Ru 0.21, Ir 0.32, Rh 0.18, Re 0.51). Second, it is not a Ru artefact — Ir loses 0.32 and Re loses 0.51. Third, the collapse is not merely novel-scaffold generalisation: holding out whole Bemis–Murcko scaffolds *within* each metal costs 0.10–0.12 for the two metals where the penalty resolves (Ru 0.10, Ir 0.12; the Rh, Os and Re cohorts are too small to resolve it), while the residual "net metal" effect (scaffold − cross) is 0.12–0.18 for Ru and Ir. Metal identity is therefore the largest single shift axis, but roughly half of

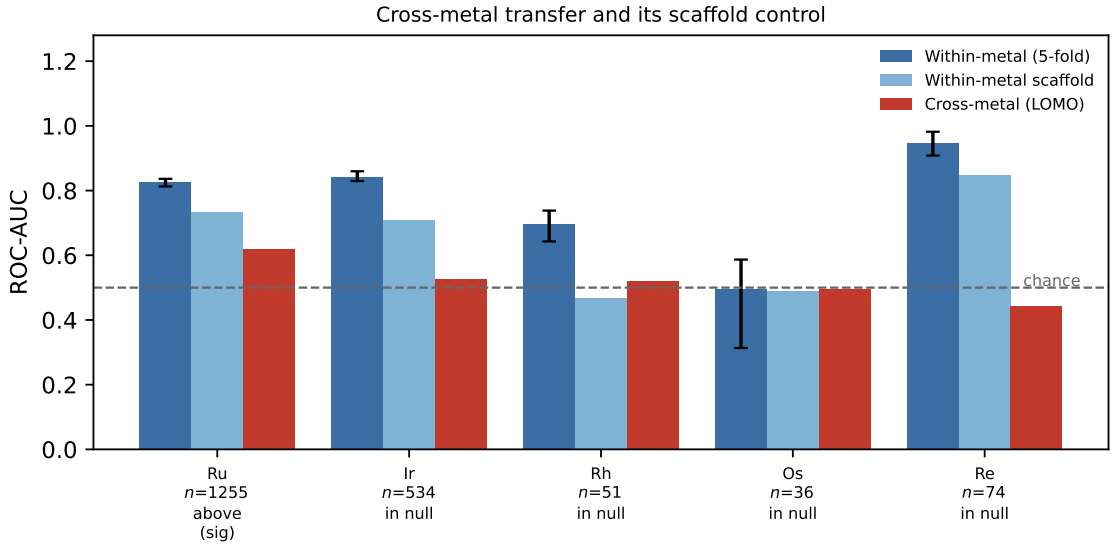



Figure 1: Transfer matrix. For each held-out metal: within-metal 5-fold AUC (5-seed mean, seed range as error bars), the same-metal Bemis–Murcko scaffold-split control, and the cross-metal LOMO AUC. The scaffold column sits between the two: metal shift is the largest single axis but not the only one. The label under each metal gives the held-out fold size and the label-permutation verdict. **MEASURED**: `metrics/by_metric/metal_activity_transfer_seed_stability.json`, `metal_activity_transfer_scaffold.json`, `metal_activity_transfer_permutation_null.json`.

the raw within-to-cross drop is ordinary out-of-distribution behaviour that any scaffold split would also incur.

Two cautions bound the per-metal reading. The held-out folds are small for Rh ($n = 51$), Os ($n = 36$) and Re ($n = 74$), and a label-permutation null on the fixed cross-metal ranking shows that none of the five metals is significantly *below* chance at these sizes — the null standard deviation is $\pm 0.09$ at $n = 51$ and $\pm 0.10$ at $n = 36$. Only Ru ($n = 1{,}255$) is significantly *above* chance *under this scorer*. We therefore report the below-0.5 point estimates for Os (0.4950) and Re (0.4429) as measured but unresolvable, not as a below-chance effect — a caution the next paragraph shows is scorer-specific for Re.

**Significance.** On the fixed seed-42 measurement the transfer penalty is distinguishable from zero for Ru (0.218, 95% CI $[0.157, 0.279]$), Ir (0.302, $[0.217, 0.380]$) and Re (0.482, $[0.231, 0.699]$), but not for Rh ($[-0.134, 0.550]$) or Os ($[-0.447, 0.425]$), whose intervals include zero. Under the label-permutation null over the five-metal family, only Ru survives a Holm–Bonferroni correction at 0.05. The two statements are not in conflict: the bootstrap interval is per-metal and uncorrected, while the permutation verdict is corrected across the family, and the three small cohorts (Rh $n = 51$, Os $n = 36$, Re $n = 74$) are exactly where the two disagree. **MEASURED**: `metrics/by_metric/metal_activity_transfer_significance.json`.

**Scorer robustness and the Re exception.** The collapse is not a fingerprint artefact. Replacing the fingerprint-plus-XGBoost scorer with the official chemprop v2 message-passing backbone (8) on the same cohort and the same ligand-disjoint LOMO split reproduces the collapse for Ru, Ir and Rh: cross-metal AUC 0.5621 $[0.5075, 0.5880]$, 0.4841 $[0.4062, 0.5033]$, 0.4915 $[0.3243, 0.6859]$

Table 2: tmQM pretraining A/B on the Ru and Ir activity heads, on the fixed Bemis–Murcko scaffold split. Paired over 5 seeds, identical split and schedule; `init` is the only variable. CIs are t-based on the 5 per-seed deltas (§4.2); a percentile bootstrap is also recorded in the artifacts but is anti-conservative at $n = 5$. **MEASURED**: `metrics/by_metric/tmqm_pretrain_init_ab_ru.json` and `tmqm_pretrain_init_ab_multimetal.json`.

| Metal | Arm | Mean ROC-AUC | Sample std over seeds |
|---|---|---|---|
| Ru ($n_{\text{test}} = 215$) | tmQM init | 0.7778 | 0.0194 |
| Ru | random init | 0.7872 | 0.0201 |
| Ir ($n_{\text{test}} = 56$) | tmQM init | 0.7205 | 0.0399 |
| Ir | random init | 0.6719 | 0.0258 |

Ru paired delta $-0.0094$, 95% CI $[-0.0406, +0.0218]$, $t(4) = -0.84$, 2/5 seeds positive
Ir paired delta $+0.0486$, 95% CI $[-0.0238, +0.1210]$, $t(4) = +1.86$, 3/5 seeds positive

(2,000-resample bootstrap of the seed-42 ranking), each at or below the fingerprint probe and at or near chance, with a within-minus-cross penalty of +0.22, +0.31 and +0.17 — the same sign as the fingerprint probe. Os is above the fingerprint probe for the backbone too (0.5559 vs 0.4950) but stays inside the null band at $n = 36$ (CI $[0.3311, 0.7078]$). Re is the exception: the graph backbone reaches cross-metal AUC 0.7368 $[0.6202, 0.8904]$ against the fingerprint probe's 0.4429 (Holm $p = 0.010$), recovering a Re signal the fingerprint loses. We therefore state Claim 1 as a collapse for Ru, Ir and Rh and report Re as representation-dependent: whether Re transfers is a property of the molecular representation, not of the metal axis alone. **MEASURED**: `metrics/by_metric/pa3_chemprop_lomo_probe_cohort.json`; the full scorer ablation, with the family-corrected verdicts, is Table 7.

The collapse is not underfitting. On the pooled cohort a constant predictor scores ROC-AUC 0.5 by construction (majority-class accuracy 0.687), but a metal-agnostic model that ignores metal identity reaches 0.8348 on a random pooled split and 0.7378 on a pooled scaffold split (`metrics/by_metric/metal_activity_transfer_baselines.json`). The scorer can learn this cohort well when the metal axis is not held out; the 0.44–0.62 cross-metal column is a distribution shift, not a failure to fit.

The Re case is the sharpest illustration. Within Re, activity is almost perfectly predictable (0.9480). Under the fingerprint scorer nothing learned from the other four metals transfers to it (0.4429); under the message-passing backbone 0.7368 of it does. Re is therefore not a metal axis no model can cross — it is a case where the fingerprint representation discards the transferable part of the signal, which is exactly the representation-dependence Claim 1 now reports. For Ru, Ir and Rh, by contrast, the test set is not a novel scaffold in familiar chemical space but a different ligand→label relationship that neither representation crosses.

## 4.2 Claim 2: no significant tmQM pretraining transfer (a two-metal honest negative)

Table 2 reports the paired A/B on two metals. The only difference between the arms is the encoder initialisation.

We report this as an honest negative, now measured on two metals rather than one. On Ru the paired delta is $-0.0094$ (t-based 95% CI $[-0.0406, +0.0218]$, $t(4) = -0.84$, 2/5 seeds positive): tmQM pretraining does not improve the Ru activity head. On Ir the point estimate is positive, +0.0486, but it is *suggestive rather than confirmed* — only 3/5 seeds are positive, the t-based 95%

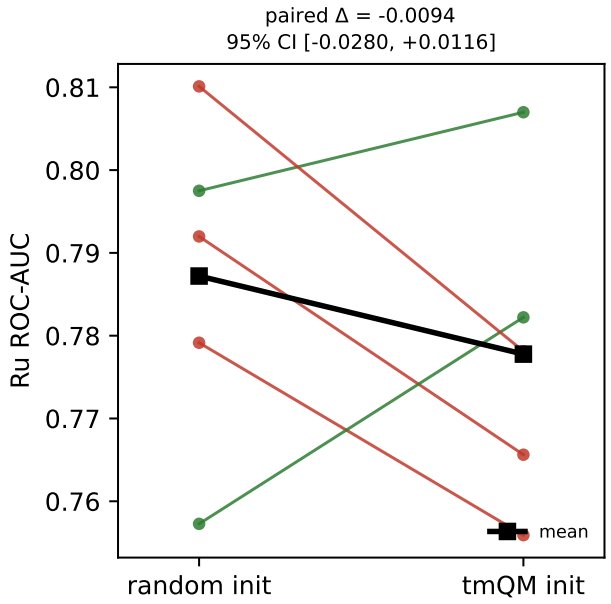



Figure 2: Paired per-seed deltas for the tmQM pretraining A/B on the Ru arm of the two-metal comparison (Table 2), on the fixed scaffold split. Two of five seeds move toward tmQM (green), three away (red); the black line is the mean. The interval crosses zero; the Ir arm is tabulated only. **MEASURED**: `metrics/by_metric/tmqm_pretrain_init_ab_ru.json`.

CI $[-0.0238, +0.1210]$ includes zero, and the paired $t(4) = +1.86$ gives a one-sided $p = 0.068$. We deliberately do not promote it. A 5-seed percentile bootstrap returns $p = 0.012$ on the same five deltas, but at $n = 5$ the bootstrap cannot generate enough distinct resamples and is anti-conservative (Wilcoxon gives $p = 0.16$); the t-test is the honest reference and Ir does not clear it. Across the two metals no Holm-corrected effect is significant, so we make no pretraining claim anywhere in this paper. The measurement itself is sound — the paired design cancels seed variance, the arms are identical apart from the initialisation, and both arms print the number of backbone tensors loaded from tmQM, so neither can be a mislabelled version of the other. What fails is the effect, not the measurement.

This reverses an earlier result, and we state the reversal rather than hide it. The same A/B measured on the degenerate splitter described in §3.2 gave +0.0181 (95% CI $[+0.0035, +0.0328]$, 5/5 seeds positive, and a 2.8× smaller standard deviation across seeds). That splitter was not scaffold-disjoint — it was molecule-level and mislabelled (§6) — and once the split is corrected the effect disappears. The superseded +0.0181 is retained in the source artifact only as the reason the protocol was tightened.

### 4.3 Claim 3: a within-metal win and a cross-cohort honest negative

Table 3 separates the two things that could explain the gap: model class and cohort. The top two rows hold the cohort and the DOI-split 10-fold partition fixed and change only the model class; the bottom row is the published model on its own (narrower) cohort.

Under the published protocol's own criteria — their split scheme, dedup rule and cell-line funnel, verified DOI-disjoint, with an md5-checked data release — our D-MPNN scores $0.7717 \pm 0.0380$, 0.0383 below the published $0.810 \pm 0.024$. But swapping in the published model class on our

Table 3: Head-to-head under the published protocol's own criteria (DOI-split 10-fold CV, shared dedup and cell-line funnel). The first two rows are *same-cohort, same-fold* and differ only in model class; the last row is the published model on its own cohort. **MEASURED**: `metrics/by_metric/metalcytotox_ru_auc.json::paper_protocol_measurement` and `metalcytotox_ru_paper_lightgbm_cv.json`.

| | ROC-AUC | Cohort (rows / complexes / cell lines) | Model class |
|---|---|---|---|
| This work | $0.7717 \pm 0.0380$ | 15,799 / 4,793 / 591 | D-MPNN |
| This work (same cohort) | $0.7677 \pm 0.0368$ | 15,799 / 4,793 / 591 | LightGBM |
| Krasnov et al. (3) | $0.810 \pm 0.024$ | 11,294 / 4,399 / 23 | LightGBM |
| Same-cohort model-class gap $-0.0040$; cross-cohort gap $-0.0383$ | | | |

identical cohort and fold partition gives $0.7677 \pm 0.0368$: a same-cohort model-class gap of only $-0.0040$, well inside either standard deviation. The model class is therefore *not* what separates us from the published number; the residual 0.0383 is the cohort scope — our funnel keeps 591 cell lines where the paper keeps 23, so we are predicting on a much broader and more heterogeneous cell-line population. That is a cross-cohort estimate, and we label it as one. The published paper reports its headline numbers on a *within-metal* axis (5-fold CV inside one metal: Ru 0.81, Ir 0.73). On that axis our fingerprint-plus-XGBoost scorer (Table 1) exceeds the published value for *Ir* under both tie rules and every seed: 0.8295–0.8597 under the current rule and 0.8237–0.8374 under the conservative rule, i.e. +0.09 to +0.13 over 0.73. We claim that win. For *Ru* we claim only parity: the interval is entirely above 0.81 under the current rule (0.8128–0.8363) and entirely at or below it under the conservative rule (0.7880–0.8082), so the direction is decided by the tie rule, not by the data. The published work reports no within-metal value for Rh, Os or Re; ours are 0.6974, 0.4960 and 0.9480 (Table 1), the first reported for those three metals. Our honest negative is therefore narrow and specific: it is the cross-cohort DOI-split *replication* that sits 0.0383 below, not the within-metal comparison.

The within-metal column is the fingerprint-plus-GBDT scorer's, and we say so explicitly. Re-measured on the *same* cohort and the *same* folds, our D-MPNN backbone scores 0.7041 (Ru) and 0.6912 (Ir) against the GBDT's 0.8271 and 0.8430 — below it on all five metals by 0.05–0.15, and also below the published 0.81/0.73. The within-metal win is therefore a property of the fingerprint-plus-GBDT scorer on this tabular-size cohort, not of the message-passing backbone: we report the column as the GBDT's and use the D-MPNN where it is the object of study (the transfer, pretraining and few-shot analyses above) rather than as the within-metal scoring baseline. **MEASURED**: `metrics/by_metric/dmpnn_within_metal_auc.json`.

Two internal controls bound the interpretation. Against our own single-task baseline on the same cohort (0.7123), the multi-task D-MPNN scores 0.7613, a +0.049 apples-to-apples gain — the architecture is not the problem. The NCI-60 panel is a weaker, *data-limited* control: on the fixed scaffold split it scores $0.7294 \pm 0.2137$ across all nine cell lines, but two of the nine have single-digit test sets (SKOV3 $n_{\text{test}} = 3$, DU145 $n_{\text{test}} = 7$) that return an uninformative AUC of 1.0; restricting to the five lines with $n_{\text{test}} \geq 20$ gives 0.5923, the number we quote for the panel. It is data-limited rather than capacity-limited: a three-seed probe that raises the encoder from 128/3 to 512/4 — the configuration that lifts MoleculeNet bace by 0.05 — moves the nine-line mean by $-0.0014$ and the five-line funnel by +0.0130, both inside seed noise ($p = 0.52$ and $p = 0.24$; the probe's own three-seed spread is 0.044). We therefore report the panel as an honest negative and make no tie claim against the 0.81 comparator (`metrics/by_metric/nci60_precious_9cell_auc.json`, `metrics/by_metric/l1_nci60_cap512_d4_probe.json`).

Table 4: Mitigating the cross-metal collapse. Top: static metal conditioning, cross-metal LOMO mean over five metals, paired against the unconditioned baseline. Bottom: few-shot adaptation, paired against the same run's zero-shot query AUC over ten seeds and five metals. Holm–Bonferroni within each family. **MEASURED**: `metrics/by_metric/pa15_metalcond_lomo.json`, `l3_metalcond_lomo.json`, `l3_kshot_curve.json`.

| Arm | Cross-metal mean | Paired $\Delta$ | Holm $p$ |
|---|---|---|---|
| *Static metal conditioning* | | | |
| Unconditioned (baseline) | 0.5693 | — | — |
| Metal token | 0.5205 | −0.0488 | 1.000 |
| Metal descriptor | 0.5480 | −0.0215 | 1.000 |
| FiLM | 0.5275 | −0.0418 | 0.984 |
| *Few-shot adaptation ($K$ labelled support ligands)* | | | |
| $K = 5$ | 0.5813 | +0.0086 | 0.286 |
| $K = 10$ | 0.6061 | +0.0306 | 0.070 |
| $K = 20$ | 0.6233 | +0.0580 | **0.0039** |
| $K = 40$ | 0.6990 | +0.0907 | **0.0036** |

### 4.4 Claim 4: static metal conditioning fails, few-shot adaptation recovers the collapse

Claim 1 shows that a ligand-only model does not transfer across metals. We ask what does. On the same cohort and the same ligand-disjoint LOMO split we test two families: *static metal conditioning* — feeding metal identity into the model, as a metal token, a metal-descriptor embedding, or a FiLM scale-and-shift on the pooled readout — and *few-shot adaptation* — freezing the zero-shot model and fine-tuning it on $K$ labelled support ligands from the held-out metal, then scoring a disjoint query set (Table 4).

Static conditioning does not work, and this is a clean negative. The metal-token and descriptor arms leave the cross-metal mean at or below the unconditioned baseline (0.5205 and 0.5480 against 0.5693), and FiLM *hurts* it (0.5275, a paired delta of −0.0418, 95% CI $[-0.0912, -0.0030]$, Holm $p = 0.984$). Knowing *which* metal is held out does not, by itself, recover the transferable signal: the collapse is not a missing-input problem.

Few-shot adaptation does work. Fine-tuning on $K$ labelled ligands from the held-out metal recovers AUC monotonically in $K$: +0.0086 at $K$=5, +0.0306 at $K$=10, +0.0580 at $K$=20 (95% CI $[+0.0237, +0.0910]$, Holm $p = 0.0039$) and +0.0907 at $K$=40 (95% CI $[+0.0379, +0.1426]$, Holm $p = 0.0036$), paired over ten seeds and five metals and significant after family correction from $K = 20$ onward. This is the paper's constructive result: the cross-metal collapse is not a dead end — a handful of labelled ligands from the target metal recovers most of it, whereas static metal conditioning recovers none. It is an *adaptation* result, so the support labels come from the held-out metal; the zero-shot LOMO column of Table 1 remains the benchmark number. **MEASURED**: `metrics/by_metric/pa15_metalcond_lomo.json`, `l3_metalcond_lomo.json`, `l3_kshot_curve.json`.

### 4.5 Supporting control: the backbone is competitive on organic benchmarks

This subsection is a *backbone control*, not a head-to-head against the paper's competitor: its only job is to show that the collapse of §4.1 is a property of the metal axis and not of an encoder that cannot fit anything. We evaluate the same D-MPNN on MoleculeNet tasks under a Bemis–Murcko scaffold split. Because the numbers usually quoted as the chemprop reference are best-of-search constants,

Table 5: Backbone control: D-MPNN and chemprop v2 on MoleculeNet tasks, identical Bemis–Murcko scaffold split, 5 seeds each. The "chemprop v2 (same protocol)" column is our own re-run under this split; "literature" is the commonly quoted best-of-search constant (2). **MEASURED**: `metrics/by_metric/moleculenet_auc_dmpnn_5seed_cap512_d4.json`, `metrics/by_metric/moleculenet_auc_chemprop_v2_same_protocol.json`.

| Task | This work (D-MPNN 512/4) | chemprop v2 (same protocol) | Δ | Literature |
|---|---|---|---|---|
| bbbp | $0.8994 \pm 0.0218$ | $0.8861 \pm 0.0135$ | +0.0133 | 0.940 |
| hiv | $0.7889 \pm 0.0189$ | $0.7807 \pm 0.0222$ | +0.0082 | 0.823 |
| bace | $0.7933 \pm 0.0250$ | $0.8458 \pm 0.0113$ | −0.0525 | 0.876 |
| clintox | $0.8303 \pm 0.0133$ | $0.8847 \pm 0.0193$ | −0.0544 | 0.894 |
| mean | 0.8280 | 0.8493 | −0.0214 | 0.8832 |

we additionally re-run the official chemprop v2 message-passing backbone (8) under the *identical* split and seeds (Table 5). Our D-MPNN scores 0.8280 (5 seeds), +0.0131 above our own earlier 128/3 configuration, and beats the protocol-matched chemprop v2 on two of the four tasks (bbbp +0.0133, hiv +0.0082). The literature constants sit 0.0339 above even a same-protocol chemprop v2 run, confirming they are best-of-search rather than protocol-matched. Seed-ensembling the five runs is a free inference-time lever that lifts the mean further on a pinned-split replication (+0.0123, §6.5); it does not change the ordering against chemprop.

Against a protocol-matched backbone we win two of the four tasks and lose two, for a 0.0214 mean gap; the two losses are generic ADMET endpoints outside this paper's target chemistry and are carried into §6.5. The point of the control is narrow and it holds: the encoder is a competent organic property predictor, so the cross-metal collapse is a property of the metal axis, not of a backbone that simply cannot fit.

# 5 Sensitivity analyses

The headline collapse (Claim 1) uses the cross-metal column, which is unaffected by the choices below. We nonetheless bound the *within-metal* column and the corpus claim, because a reader is entitled to know where a conclusion is decided by a preprocessing rule rather than by the data.

## 5.1 The tie rule decides the Ru comparison

When duplicate measurements of the same ligand disagree, the collapse rule resolves an exact tie toward *active* (predicate sum $\times$ 2 $\geq n$). This is the anti-conservative direction for an activity claim, and it is the rule both the fingerprint probe and the neural scorer use, so the within/cross comparison is internally consistent. But 80 ligands (4.1% of the deduplicated cohort) are exact ties, so it is worth asking how much of the headline depends on the choice (Table 6).

The consequence is asymmetric and we state it plainly. The *Ir* claim is robust: Ir exceeds 0.73 under both rules and every seed. The *Ru* claim is *not* robust: under the conservative rule Ru sits at 0.7880–0.8082, entirely at or just below 0.81; under the current rule it sits at 0.8128–0.8363, entirely above. Whether Ru "beats" the published value is decided by the tie rule, not by the chemistry. We therefore write Ru as *at parity* with 0.81, and do not claim a win. The collapse itself is unaffected, because it is a difference of two columns computed under the same rule.

Table 6: Within-metal AUC under the two tie rules. **MEASURED**: `metrics/by_metric/metal_activity_transfer_tie_sensitivity.json`.

| | Ties → active (current) | Ties → inactive (conservative) |
|---|---|---|
| Ru | 0.8128–0.8363 | 0.7880–0.8082 |
| Ir | 0.8295–0.8597 | 0.8237–0.8374 |
| Krasnov published: Ru 0.81, Ir 0.73 | | |

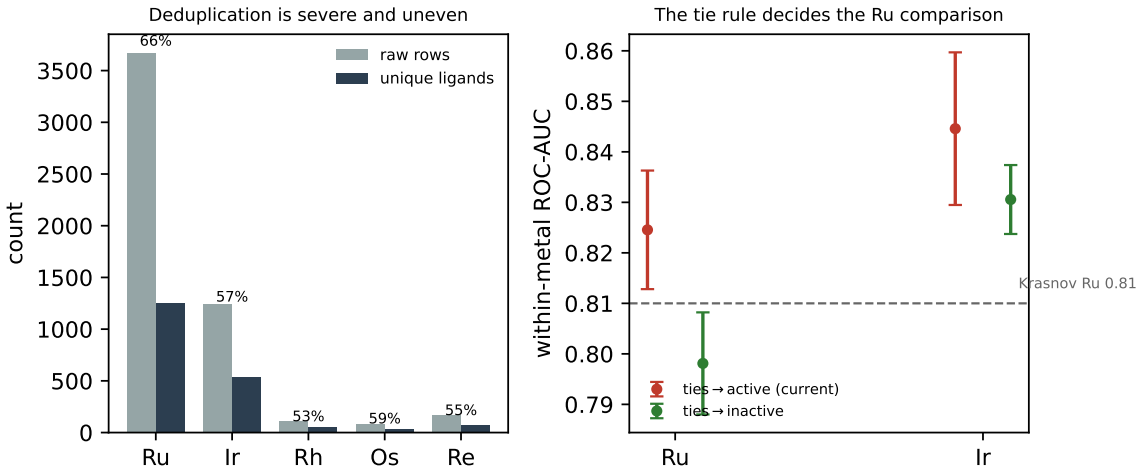



Figure 3: Left: raw rows versus unique ligands per metal (percentage is the duplication rate). Right: within-metal AUC interval across seeds under the two tie rules; the Ru interval crosses the published 0.81 line, the Ir interval does not. **MEASURED**: `metrics/by_metric/metal_activity_transfer_seed_stability.json`, `metal_activity_transfer_tie_sensitivity.json`.

## 5.2 Deduplication

Raw MetalCytoToxDB rows are heavily duplicated at the ligand level: Ru's 3,668 rows collapse to 1,255 unique ligands (65.8% duplication), and the duplication is uneven across metals. Both the within-metal and cross-metal numbers are computed on the deduplicated cohort, so the two columns are comparable to each other. Deduplication matters because the same ligand can be measured against several metals; without it, a row split would put a ligand in both train and test and inflate both columns. The LOMO protocol removes this leak directly by dropping any training ligand that appears in the held-out metal, so the cross-metal column is ligand-disjoint by construction.

## 5.3 Scorer choice

Claim 1's cross-metal column is measured with a Morgan-fingerprint plus XGBoost probe. To show the collapse is not a fingerprint artefact we re-run the ligand-disjoint LOMO evaluation with the official chemprop v2 message-passing backbone (8) on the same cohort and the same splits (Table 7).

The backbone reproduces the collapse on Ru, Ir and Rh (cross AUC 0.5621, 0.4841, 0.4915),

Table 7: Scorer ablation on the Paper-A probe cohort: same cohort, same ligand-disjoint LOMO split. Cross AUCs are 5-seed means for the backbone and a single seed (42) for the fingerprint probe; the interval is a 2,000-resample bootstrap of the backbone's seed-42 cross ranking. The verdict is a label-permutation null on that ranking, Holm–Bonferroni-corrected over the five-metal family. **MEASURED**: `metrics/by_metric/pa3_chemprop_lomo_probe_cohort.json`.

| Metal | $n_{\text{test}}$ | Fingerprint | Backbone | Backbone 95% CI | Holm $p$ |
|---|---|---|---|---|---|
| Ru | 1,255 | 0.6179 | 0.5621 | $[0.5075, 0.5880]$ | 0.040 |
| Ir | 534 | 0.5276 | 0.4841 | $[0.4062, 0.5033]$ | 0.228 |
| Rh | 51 | 0.5190 | 0.4915 | $[0.3243, 0.6859]$ | 1.000 |
| Os | 36 | 0.4950 | 0.5559 | $[0.3311, 0.7078]$ | 1.000 |
| Re | 74 | 0.4429 | 0.7368 | $[0.6202, 0.8904]$ | 0.010 |

Table 8: Bond perception on 900 held-out tmQM complexes, identical cohort and per-complex aggregation for every column. The learned head and OpenBabel are scored on non-hydrogen pairs; RDKit (stripped) is scored on its native non-metal *and* non-hydrogen pair scope. The head column is a mean ± sample std over three full-corpus checkpoints (seeds 0–2); OpenBabel and RDKit are deterministic given the input, so their only uncertainty is the sampling of the 900 complexes. **MEASURED**: `metrics/by_metric/tmqm_bond_perception_3seed.json`.

| | Learned head | OpenBabel (40) | RDKit (stripped) (41) |
|---|---|---|---|
| bond EXISTENCE F1 | $\mathbf{0.9962 \pm 0.0008}$ | 0.9771 | 0.9998 |
| exact-type rate | $\mathbf{0.0956 \pm 0.0113}$ | 0.0300 | 0.0220 |
| macro-F1(type) | $0.4784 \pm 0.0055$ | **0.5051** | 0.2809 |
| success rate | 1.000 | 1.000 | 0.303 |

keeps Os inside its null band, and recovers Re (0.7368). Under the family correction only Re and Ru survive: Ir's below-chance reading under the backbone is significant only uncorrected (one-sided $p = 0.038$; two-sided 0.076, Holm 0.23). The collapse for Ru, Ir and Rh is therefore robust to the scorer, while the Re exception is a property of the representation, not of the metal axis. The within-metal column carries no bootstrap interval here because the PA3 report stores per-fold AUCs but not the per-ligand out-of-fold scores a within-metal bootstrap needs; the fingerprint within column's interval is in §4.1.

### 5.4 tmQM bond-perception reference

The tmQM corpus is the pretraining source for Claim 2 and, in a companion line of work, the ground truth for metal bond-order perception. For completeness we report the same-protocol bond-perception comparison on 900 held-out complexes (150 per metal, non-hydrogen pair scope, per-complex aggregation, Table 8).

The learned head wins two of three axes, including a 3.2× edge in exact-type rate, but it *loses* the axis it was designed to win: macro-F1(type) $0.4784 \pm 0.0055$ against OpenBabel's 0.5051, 0.0267 behind. The loss is consistent across all three seeds and is nearly five times the between-seed standard deviation, so it is not seed noise. We report the losing row deliberately. We also record a withdrawn number: an earlier figure of 0.9194 came from a *pooled* aggregation on a different cohort, and the seed-0 checkpoint scores 0.4723 under the per-complex protocol used here. The two are not interchangeable, and quoting 0.9194 beside OpenBabel's per-complex 0.5051 would compare

three things at once.

# 6 Limitations and honest negatives

We separate what the study establishes from what it does not.

## 6.1 Cohort scope and the cross-cohort caveat

The head-to-head in Table 3 is a *cross-cohort* estimate. Our funnel keeps 591 cell lines against the paper's 23; we adopt their split scheme and dedup rule exactly, but the underlying population differs. We can now attribute the 0.0383 gap: running the published model class on our identical cohort and fold partition closes it to $-0.0040$, so the gap is cohort scope, not model class. The claim we defend is therefore: under the published protocol, on a broader and more heterogeneous cohort, a D-MPNN and the published LightGBM are statistically indistinguishable, and both sit 0.038 below the number the paper reports on its narrower cohort.

## 6.2 Small per-metal cohorts make part of the result unresolvable

Three of the five metals have too few unique ligands to resolve an effect: Rh 51, Re 74, Os 36. Os is the clearest case: its within-metal AUC swings from 0.3133 to 0.5867 across five seeds (spread 0.2734), larger than any method change would move it. That swing *is* the unresolvability. We therefore report Os, Rh and Re as measured-but-not-gated and draw no metal-level conclusion from them individually. In particular, the sub-0.5 cross-metal point estimates for Os (0.4950) and Re (0.4429) are *not* a below-chance effect: a label-permutation null on the fixed cross-metal ranking places both inside the null (null SD $\pm 0.10$ at $n = 36$, $\pm 0.09$ at $n = 74$), and only Ru is significantly above chance under the fingerprint scorer after Holm correction (the message-passing backbone places Re above chance at 0.7368, Holm $p = 0.010$; see §4.1). Re's three-seed cross-metal range (0.4429–0.4798) is tight, but a tight range around an unresolvable level is still unresolvable.

## 6.3 The pooled cross-metal mean is Ru-dominated

Ru is 64% of the deduplicated cohort. Any pooled cross-metal mean is largely a Ru number, and quoting it without the per-metal table would overstate the breadth of the finding. We quote per-metal throughout and treat the pooled value as descriptive only.

## 6.4 A cohort-construction trap in a related protocol

A preprocessing filter applied to a pre-filtered artifact can manufacture a below-chance result, so we record one we hit while building a related leave-one-metal-out protocol. The default cohort file is a snapshot already filtered to short exposure times ($\leq 24$ h); raising the threshold at training time cannot recover the dropped rows, and the loaded cohort stayed byte-identical — the filter silently did nothing. The consequence is a below-chance fold: the Rh held-out set collapses to 108 test ligands and scores 0.3812, dragging the pooled LOMO mean to 0.5575, with no change to the model. Readers building on MetalCytoToxDB can hit the same trap. **MEASURED**: `metrics/by_metric/metal_lomo_auc_wasserstein.json`.

### 6.5 Two MoleculeNet task gaps remain

Under a protocol-matched comparison (our own chemprop v2 re-run, same scaffold split and seeds) the D-MPNN wins bbbp (+0.0133) and hiv (+0.0082) and loses bace ($0.7933 \pm 0.0250$ against $0.8458 \pm 0.0113$, $-0.0525$) and clintox (0.8303 against 0.8847, $-0.0544$). Both losses exceed seed noise, so we report them as honest negatives. They are generic ADMET endpoints outside this paper's target chemistry and do not bear on the metal-axis claims; the backbone is otherwise competitive (mean 0.8280), which is what lets us argue in §4.5 that the metal collapse is a property of the axis and not of a broken model. Seed-ensembling the five runs is a free inference-time lever that lifts the mean by +0.0123 on a pinned-split replication (`l2_moleculenet_ensemble.json`) without closing the bace gap.

### 6.6 Leakage controls

Cross-metal evaluation is only meaningful if test ligands are absent from training. The LOMO protocol drops any training ligand that appears in the held-out metal, so the cross-metal column is ligand-disjoint by construction. The within-metal column uses 5-fold CV over a metal's own ligands and shares no ligands across folds by construction. The scaffold control in Table 1 holds out whole Bemis–Murcko scaffolds within a metal, so its "net metal" column is the collapse with the ordinary novel-scaffold penalty already removed. Two caveats remain: the scaffold-split control is a single seed (42), and both the cross-metal and scaffold splits are ligand-/scaffold-disjoint but not cell-line-disjoint.

### 6.7 What we do not claim

- We do not claim the collapse is caused by any specific chemical mechanism. We measure that it happens; we do not identify why.
- We do not claim any metal's cross-metal AUC is *below* chance under the fingerprint scorer. Under its label-permutation null none is, and only Ru is significantly above chance after Holm correction. The message-passing backbone moves Re above chance (0.7368, Holm $p = 0.010$); Ir's below-chance reading is significant only as an uncorrected one-sided test ($p = 0.038$, two-sided 0.076, Holm 0.23) and does not survive the family correction, so we do not claim it. The sub-0.5 fingerprint point estimates for Os and Re are unresolvable at $n = 36$–$74$.
- We do not claim metal is the *only* shift axis. A same-metal scaffold split already costs 0.10–0.12 on the two metals where the penalty resolves; the metal axis is the largest single remaining shift, not the sole source of the collapse.
- We do not claim to beat the published model on its DOI-split replication: we are 0.0383 below it cross-cohort, and the same-cohort model-class gap is $-0.0040$. We *do* claim a within-metal win for Ir with the fingerprint-plus-XGBoost scorer (Table 1: +0.09 to +0.13, robust to both tie rules and every seed); Ru is at parity, decided by the tie rule. The within-metal column is that scorer's, not the D-MPNN's: re-measured on the same cohort and folds the D-MPNN scores 0.7041 (Ru) and 0.6912 (Ir) against the GBDT's 0.8271 and 0.8430, i.e. below on all five metals.
- We do not claim a tmQM pretraining benefit. On the fixed scaffold split, measured on two metals, no Holm-corrected effect is significant: Ru's paired delta is $-0.0094$ (t-based 95% CI $[-0.0406, +0.0218]$), and Ir's is a suggestive +0.0486 that does not clear the paired $t$-test (3/5

seeds positive, CI includes zero, one-sided $p = 0.068$; the 5-seed bootstrap's $p = 0.012$ is anti-conservative at that sample size). The earlier +0.0181 came from a degenerate, molecule-level splitter and does not survive the fix.

- We do not claim the bond-perception head wins. It loses macro-F1(type).
- We do not claim Ru exceeds the published 0.81 unqualified; that is decided by the tie rule (§5.1).
- We do not claim static metal conditioning mitigates the collapse. A metal token, a metal-descriptor embedding and FiLM all leave the cross-metal mean at or below the unconditioned baseline (§4.4); the mitigation that works is few-shot adaptation to the target metal, and it needs 20–40 labelled ligands.

# 7 Conclusion

Metal identity is a first-class distribution shift for ligand-based activity models. On a deduplicated MetalCytoToxDB cohort, a single shared scorer loses 0.18–0.51 ROC-AUC when it moves from a metal it has seen to one it has not, and a same-metal scaffold-split control shows that only about half of that drop is ordinary novel-scaffold generalisation. Under a Holm-corrected label-permutation null, and at the cohort sizes available, the cross-metal signal is significantly above chance only for Ru under the fingerprint scorer; an independent message-passing backbone additionally recovers Re (0.7368, Holm $p = 0.010$), which is why we report Re as representation-dependent rather than as a failed transfer. The sub-0.5 fingerprint point estimates for Os and Re are not below-chance effects. tmQM pre-training of the encoder does *not* transfer to the activity head on a genuine scaffold split, on either of the two metals tested (Ru −0.0094; Ir a suggestive +0.0486 that does not clear the paired $t$-test, $p = 0.068$), and running the published protocol's own model class on our identical cohort and folds reproduces it to within −0.0040 — so the honest negative is about cohort scope, not model class.

The actionable reading is narrow but clear. A ligand-only QSAR model trained on one metal should not be deployed to rank candidates for a different metal without re-validation: the features that carry within-metal signal do not survive the metal change, and feeding the model the metal identity does not repair this — static metal conditioning (token, descriptor, FiLM) leaves the cross-metal mean at or below the unconditioned baseline. What does work is cheap: a handful of labelled ligands from the target metal (20–40) recovers most of the collapse (+0.058 to +0.091 AUC, Holm $p < 0.004$). We therefore recommend (i) few-shot adaptation to the target metal rather than a metal-conditioned zero-shot model, and (ii) reporting cross-metal leave-one-metal-out alongside the within-metal number — with a same-metal scaffold split as the control — whenever a dataset spans more than one metal. Cross-metal transfer deserves to be a reported benchmark axis, not an implicit assumption.

# Data and code availability

All cohorts, configs and scripts are versioned in the accompanying repository. Every quantitative claim in this paper cites a JSON artifact under `metrics/by_metric/`, listed at the corresponding table or figure.

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